

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause change of state)	B1
	(thermal) energy to change state at constant temperature	B1
3(b)(i)	$W = p\Delta V$	C1
	$= 1.0 \times 10^5 \times 0.017 = 1700 \text{ J} = 1.7 \text{ kJ}$	A1
3(b)(ii)	$\Delta U = Q + W$	C1
	$Q = 17.6 + 1.7$ $= 19.3 \text{ kJ}$	A1
3(b)(iii)	mass $= 710 \times 7.2 \times 10^{-5}$ (= 0.051 kg)	C1
	$L = 19.3 / 0.051$ $= 380 \text{ kJ kg}^{-1}$	A1
3(c)	fusion involves (much) smaller volume change (than vaporisation)	B1
	smaller change in intermolecular spacing so smaller change in internal energy	B1
	negligible work done (by substance during fusion) so L_F is less (than L_V)	B1

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